

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Database Management Systems | Course Code: 21CS53 | Credits: 4

Exam Session: CIA-1 Midterm 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

- Q1.** Given relation $R(A, B, C, D, E)$ and Functional Dependencies $F = \{ A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A \}$. (i) Find Candidate Keys of R , (ii) Determine highest normal form of R , and (iii) Decompose into BCNF with Lossless Join property. [8 Marks]

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- Q2.** Explain Conflict Serializability. Test if schedule $S: r1(X); r2(Y); w1(X); r2(X); w2(Y); w2(X)$ is conflict serializable by drawing the Precedence Graph. [8 Marks]